

Discrete Mathematics

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Chapter 1

Logic

1.1 Propositional Logic

Definition 1.1.1 (Proposition). A **proposition** is a statement with precise meaning which have a truth value **true** or **false**.

Definition 1.1.2 (Operators). An **operator** or **connective** combines one or more operand expressions into a larger expression.

Table 1.1: Operator of propositions (Ordered by Precedence)

Formal Name	Nickname	Arity	Symbol	Precedence
Negation operator	NOT	Unary	$\neg$	1
Conjunction operator	AND	Binary	$\wedge$	2
Exclusive disjunction operator	XOR	Binary	$\oplus$	3
Disjunction operator	OR	Binary	$\vee$	4
Implication operator	IMPLIES	Binary	$\rightarrow$	5
Biconditional operator	IFF	Binary	$\leftrightarrow$	6

Definition 1.1.3 (Converse, Inverse, and Contrapositive). The three statements of

an implication are defined as followed:

Converse: $q \rightarrow p$

Inverse: $\neg p \rightarrow \neg q$

Contrapositive: $\neg q \rightarrow p$.

The meaning of the contrapositive of $\neg q \rightarrow \neg p$ is same as $p \rightarrow q$. Note that the “positive” in “contrapositive” actually means “position” in English.

Example 1.1.4. $p \vee (\neg p \wedge q) \equiv (p \vee \neg p) \wedge (p \vee q) \equiv T \wedge (p \vee q) \equiv p \vee q$.

Definition 1.1.5 (Equivalence). Compound proposition p is logically equivalent to compound proposition q , written $p \equiv q$, if the compound proposition $p \leftrightarrow q$ is a tautology.

Definition 1.1.6 (Tautology). A tautology is a compound proposition that is always true.

For example, $p \vee \neg p$ is a tautology.

Definition 1.1.7 (Contradiction). A contradiction is a compound proposition that is always false.

For example, $p \wedge \neg p$ is a contradiction.

Some equivalence laws are listed as followed:

Theorem 1.1.8 (Some Logical Equivalences).

1. Identity: $p \wedge T \equiv p$ $p \vee F \equiv p$.
2. Domination: $p \wedge F \equiv F$ $p \vee T \equiv T$.
3. Idempotent: $p \wedge p \equiv p$ $p \vee p \equiv p$.
4. Double negation: $\neg \neg p \equiv p$.
5. Associative: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$. $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
6. Distribution: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Theorem 1.1.9 (Logical Equivalences Involving Conditional Statements).

1. $p \rightarrow q \equiv \neg p \vee q$.
2. $p \rightarrow q \equiv \neg q \rightarrow \neg p$.
3. $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$.
4. $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$.
5. $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$.

Theorem 1.1.10 (Logical Equivalences Involving Biconditional Statements).

1. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.
2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$.
3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.
4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$.
5. $p \leftrightarrow q \equiv \neg(p \oplus q)$.

Theorem 1.1.11 (Logical Equivalences Involving Exclusive Disjunction).

1. $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q) \equiv \neg(p \leftrightarrow q)$.

1.2 Predicate Logic

Definition 1.2.1 (Predicate logic). A **predicate logic** is an extension of propositional logic that permits concisely reasoning about whole classes of entities.

For example, “For all $x \in \{1, 2, 3\}, x < 5$ ” is a proposition with x as the **object**, $\{1, 2, 3\}$ as the **universess of discourse**, and $x < 5$ as the **predicate** or the **propositional function** mapping $x \rightarrow \{T, F\}$. Furthermore, “For all” is a kind of **quantifier**, which is defined as followed.

Chapter 2

Proofs

2.1 Inference and Introduction

Definition 2.1.1 (Proofs). A **proof** is a correct(well-reasoned, logically valid) and complete(clear, detailed) argument that rigorously and undeniably establishes the truth of a mathematical statement.

Definition 2.1.2 (Theorems). A **theorem** is a statement that has been proven to be true.

Definition 2.1.3 (Axioms, Postulates, Hypotheses, Premises). An **axiom, postulate, hypotheses, or premises** is an assumption defining the structure about which we are reasoning.

Definition 2.1.4 (Rules of Inference). A **rule of inference** is a pattern of logically valid deductions from hypotheses to conclusions.

Definition 2.1.5 (Lemmas). A **lemma** is a minor theorem used as stepping-stone to proving a major theorem.

Definition 2.1.6 (Corollaries). A **corollary** is a minor theorem proved as an easy consequence of major theorem.

Definition 2.1.7 (Conjectures). A **conjecture** is a statement whose truth value has not been proven.

Definition 2.1.8 (Theories). A **theory** is a set of all theorems that can be proven from a given set of axioms.

Now we talk about **inference rules** to help us construct proofs. Inference rules are tautologies allows us to proof the consequence true by antecedents, that is

$$\text{antecedent 1} \wedge \text{antecedent 2} \wedge \dots \rightarrow \text{consequence},$$

or we write

$$\frac{\begin{array}{c} \text{antecedent 1} \\ \text{antecedent 2} \dots \end{array}}{\therefore \text{consequence}}$$

Here are several rules:

1. Rule of addition

$$\frac{p}{\therefore p \vee q}$$

2. Rule of simplification

$$\frac{p \wedge q}{\therefore p}$$

3. Rule of conjunction

$$\frac{\begin{array}{c} p \\ q \end{array}}{\therefore p \wedge q}$$

4. Rule of modus ponens

$$\frac{\begin{array}{c} p \\ p \rightarrow q \end{array}}{\therefore q}$$

5. Rule of modus tollens (This kind of proof is called **indirect/ contrapositive proof**, example: a is rational, b is irrational, prove that $a + b$ is irrational.)

$$\frac{\neg q}{\frac{p \rightarrow q}{\therefore \neg p}}$$

6. Rule of hypothetical syllogism

$$\frac{p \rightarrow q}{\frac{q \rightarrow r}{\therefore p \rightarrow r}}$$

7. Rule of disjunctive syllogism

$$\frac{p \vee q}{\frac{\neg p}{\therefore q}}$$

Another proof similar to indirect proof is **proof by contradiction**, that is, to show that p is true, we assume $\neg p$ is true, and then find a tautology that is $\neg p \rightarrow (q \wedge \neg q) \equiv F$, then we know that p must be true. (For example, the proof of “ $\sqrt{2}$ is irrational”)

2.2 Methods and Strategy

A proof of a statement of the form $\exists x P(x)$ is called an **existence proof**. If the proof demonstrate how actually find or construct a specific element a such that $P(a)$ is true, then it is called a **constructive proof**. Otherwise, it is called a **nonconstructive proof**.

Example 2.2.1 (An Constructive Proof). Prove that for any integer $n > 0$, there exists a sequence of n consecutive composite integers.

Proof. Consider $(n+1)!+i+1$ for $i = 1, 2, 3, \dots, n$. We know that $i+1|(n+1)!+i$. $\square$

Example 2.2.2 (An Non-Constructive Proof). Prove tha there are infinitely many prime numbers.

Proof. We show that $\forall n \exists p (p > n) \wedge (p \text{ is prime})$. Consider $n! + 1$, since $n! + 1 > 1$, so $n! + 1$ is either prime or composite. Using proof by case, if $n! + 1$ is prime, then we find $p = n! + 1 > n$. Otherwise, if $n! + 1$ is composite, it must have prime factor p . If $p \leq n$, then $n! + 1 \bmod p = 1$. A contradiction. Therefore, we have $p > n$. $\square$

Chapter 3

Set Theory

3.1 Definition and Relations

Definition 3.1.1 (Sets). A **set** is a type of structure representing an unordered collection of zero or more distinct objects.

There are two kinds of notation representing sets. First, we can denote a set S by listing all of its elements in curly braces. For example $\{1, 2, 3\}$. Note that $\{1, 1, 2, 3\}$ and $\{3, 2, 1\}$ have the same meaning because elements of a set is unordered and distinct. Second, a set in the set builder notation represents a set of all elements satisfying a proposition over a domain. For example, $\{x|x > 0\}$ over $\mathbb{R}$.

Definition 3.1.2 (Equality of Sets). Two sets are declared to be **equal** if and only if they contain exactly the same elements.

Some symbols represents special infinite sets, including

1. Natural numbers: $\mathbb{N} = \{0, 1, 2, \dots\}$.
2. Integers: $\mathbb{Z}$.
3. Real numbers: $\mathbb{R}$.
4. Empty set, or null: $\emptyset$

Definition 3.1.3 (The Empty Set, or Null). $\emptyset$ represents the **empty set**, or **null**, which is the unique set that contains no elements.

Note that $\emptyset = \{\} = \{x|F\}$. Now let's look about relations. A basic set relations is **member of**. We write $x \in S$ if x is an element or member of set S . Now we can define set equality in terms of $\in$ relation:

$$\forall S, T : (S = T) \leftrightarrow (\forall x : x \in S \leftrightarrow x \in T).$$

Also, we have

$$\neg \exists x : x \in \emptyset.$$

Definition 3.1.4 (Subsets, Supersets). Let S, T be two sets. If $\forall x : x \in S \rightarrow x \in T$, we say that S is a **subset** of T , denoted $S \subseteq T$, and T is a **superset** of S , denoted $T \supseteq S$. Furthermore, $\not\subseteq$ and $\not\supseteq$ denotes the negative statements.

Theorem 3.1.5. For any set S , $S \subseteq S$.

Proof. $\forall x : x \in S \rightarrow x \in S$ is true. □

Theorem 3.1.6. For any set S , $\emptyset \subseteq S$.

Proof. Since $\forall x : x \in \emptyset$ is false, therefore, $\forall x : x \in \emptyset \rightarrow x \in S$ is true. □

Definition 3.1.7 (Strict/Proper Subsets and Supersets). Let S, T be two sets. If $S \subseteq T \wedge S \neq T$, we say that S is a **proper/strict subset** of T , denoted $S \subset T$, and T is a **proper/strict superset** of S , denoted $T \supset S$. Furthermore, $\not\subset$ and $\not\supset$ denotes the negative statements.

Definition 3.1.8 (Cardinality). The **cardinality** of a set S , denoted $|S|$, is a measure of how many different elements S has.

For example, $|\emptyset| = 0$, $| \{1, 2\} | = 2$. If $|S| \in \mathbb{N}$, we say that S is **finite**, otherwise, we say that S is **infinite**.

Definition 3.1.9 (Power Sets). Let S be a set, the **power set** of S

$$P(S) := \{x | x \subset S\}.$$